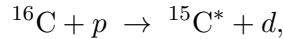


A Student's Guide to the $^{16}\text{C}(p, d)^{15}\text{C}$ Analysis

From raw data to the ^{15}C spectrum (a1975, AT-TPC, genfit)

1 What you are measuring

The a1975 experiment is a ^{16}C beam (~ 192 MeV, ~ 12 MeV/u) stopped in an **H₂ gas active target** at 1 bar; the gas is *both* the proton target and the tracking medium. You will analyze the neutron-pickup reaction



i.e. $^{16}\text{C}(p, d)^{15}\text{C}$, by detecting the outgoing **deuteron**. From the deuteron's kinetic energy T_d and lab angle θ_{lab} you reconstruct the ^{15}C excitation energy E_x with two-body relativistic kinematics, and you study which ^{15}C states are populated versus angle.

This guide goes **from scratch**: raw HDF5 $\rightarrow$ reconstruction $\rightarrow$ deuteron fit $\rightarrow$ excitation-energy spectrum, using only the macros in this directory.

2 Setup and data layout

Run everything from the (p,d) macro directory after sourcing the framework environment:

```
cd <ATTPCROOT>/macro/Unpack_HDF5/a1975/UKF
source <ATTPCROOT>/build/config.sh
```

Choose *one* base directory for the data and export it; everything below refers to it as **\$DATA**. Substitute your own path (e.g. on the analysis machine it is `/mnt/f/a1975`):

```
export DATA=/path/to/a1975          # <-- set this to where your data lives
```

The expected layout under **\$DATA** is:

- **\$DATA/h5/** raw HDF5, one file per run: `run_0106.h5`, ..., `run_0189.h5`
- **\$DATA/reco/** you will produce `*reco.root` here (and the experiment's `*FRIB.root` ion-chamber files live here)
- **\$DATA/reco_pd/** your (p,d) fit outputs (`mkdir -p $DATA/reco_pd`)

The macros are in `pp/` and `pipeline/`. Static macros write a PNG (open with your image viewer); the interactive ones (`explore_pd`, `pid_gate_edit`) need an X display, so run them with `root -l` (no `-b`).

One path to set in the macro. The reconstruction macro reads its raw input from a directory defined near the top of `pipeline/unpackReco_a1975_UKF.C` (the line `TString filepath = "..."`). Edit it once to point at **\$DATA/h5/**. Every later macro takes its directories as *arguments*, so you pass **\$DATA/...** directly.

3 The pipeline

```
$DATA/h5 → [unpack + PSA + space-charge + PRA] → $DATA/reco/*_reco.root →  
[deuteron genfit fit] → $DATA/reco_pd/*_genfitter_pd.root → [cache] → /tmp/pd_kin.root →  
[plots]
```

Pattern recognition (finding tracks) does not depend on the assumed particle, so one reconstruction serves every channel; for (p,d) you then *fit* the tracks as deuterons.

4 Step 1 — Reconstruct from raw data

`unpackReco_a1975_UKF.C` runs the full chain (unpack → pulse-shape analysis → space-charge correction → pattern recognition) and writes one `*_reco.root` per run (containing the corrected hits and the found tracks `AtPatternEvent`). For one run:

```
root -l -b -q 'pipeline/unpackReco_a1975_UKF.C("run_0106", -1, false, "'$DATA'/reco/")'
```

Arguments: (run, nEvents=-1 for all, persistRaw, outDir). This is the slow stage; run the whole experiment on 4 cores:

```
ls $DATA/h5/run_*.h5 | grep -oE 'run_[0-9]+' | sort -u > /tmp/runs.txt  
cat /tmp/runs.txt | xargs -P4 -I{} \  
    root -l -b -q "pipeline/unpackReco_a1975_UKF.C(\"{}\", -1, false, \"$DATA/reco/\")"
```

5 Step 2 — Fit the deuterons

`fitGenfitter_a1975.C` reads a `*_reco.root`, keeps only tracks inside the deuteron PID band (`pid/deuteron_band.json`), fits each as a deuteron with genfit, and writes the fitted kinematics. Its defaults are already the deuteron hypothesis:

```
cat /tmp/runs.txt | xargs -P4 -I{} \  
    root -l -b -q "pipeline/fitGenfitter_a1975.C(\"{}\", -1, \"$DATA/reco/\", \"_pd\", \"  
    $DATA/reco_pd/\")"
```

Arguments: (run, nEvents, inputDir, outputSuffix, outputDir), output `$DATA/reco_pd/run_XXXX.genfit`. A deuteron is rare (~1.5% of tracks), so most events print `0/1 tracks fitted` — that is the gate working, not an error. Expect ~30,000 deuterons over the 84 runs.

6 Step 3 — Cache the kinematics

Cache just the per-deuteron numbers into a flat ntuple (also pulls the **ion-chamber** value from the `*_FRIB.root` files for a ^{16}C beam gate):

```
cat /tmp/runs.txt | xargs -P4 -I{} \  
    root -b -q "pp/cache_pd_run.C(\"{}\", \"/tmp/pdkin_{}.root\", \"$DATA/reco_pd/\", \"  
    $DATA/reco/\")"  
hadd -f /tmp/pd_kin.root /tmp/pdkin_run_01*.root
```

This makes `/tmp/pd_kin.root` (ntuple pk: ke, theta, vz, chi2ndf, ic, run). All later plots read this single file, so they are fast.

7 Step 4 — The excitation-energy spectrum

From T_d and θ_{lab} the macros compute $E_x(^{15}\text{C})$ with two-body kinematics. Make the 4-panel overview:

```
root -l -b -q 'pp/plot_pd.C' # -> /tmp/pd_spectrum.png
```

You should see a sharp ^{15}C ground-state peak near $E_x = 0$, excited states ($\sim 3.10, 4.66$ MeV), and the deuteron T_d -vs- θ_{lab} band lying on the analytic kinematic line. A χ^2/ndf cut and the IC beam gate (950–1350) are applied by default.

8 Step 5 — Angular dependence

A discrete state appears at the same E_x at every angle, with an angle-dependent strength. Plot E_x in θ_{cm} slices:

```
root -l -b -q 'pp/plot_pd_angbins.C(10,5,12)' # start 10 deg, 5 deg wide, 12 bins
```

The ground-state resolution is best forward ($\sigma \approx 0.15$ MeV at 20–30°).

9 Step 6 — Correct the E_x -vs- θ_{cm} drift (important!)

The reconstructed ground state is not flat in θ_{cm} — its centroid drifts by ~ 0.9 MeV, which smears the integrated spectrum. Measure and subtract that drift:

```
root -l -b -q 'pp/excorr_pd.C' # -> /tmp/pd_excorr.png
```

Left: the measured drift and the curve subtracted. Right: integrated ^{15}C spectrum before/after — the g.s. FWHM improves from ~ 1.7 MeV to ~ 0.43 MeV. This is the single most important correction for the integrated spectrum; it is built into `plot_pd_angbins.C` (arg `driftCorr`, on by default) and the explorer (a checkbox).

10 Step 7 — Beware triton contamination

$^{16}\text{C}(p,t)^{14}\text{C}$ makes **tritons**. The deuteron PID gate is loose enough to admit some, and genfit fits a triton *as a deuteron* with a good χ^2 , so you cannot remove them with a χ^2 or IC cut — they form a faint second band above the deuteron locus. Inspect the PID plane and tighten the gate:

```
root -l -b -q "pp/pid_pd.C(106,113,\"$DATA/reco_pd/\")" # sqrt(dE/dx) vs Brho + gate
root -l 'pp/pid_gate_edit.C' # draw a tighter gate, Save JSON, re-run Step 2
```

Species separation must come from the **gate**, not the fit.

11 Step 8 — Interactive exploration

For tuning cuts/binning live (needs a display):

```
root -l 'pp/explore_pd.C'
```

Controls: beam energy, χ^2/ndf , θ window, IC window, vertex- z bounds, E_x binning, and checkboxes for the drift correction and the ^{15}C reference lines.

12 Physics checklist

- ^{15}C g.s. ($1/2^+$) at $E_x = 0$; first excited state 0.740 MeV ($5/2^+$, on the g.s. shoulder).
- Prominent strength near 3.10 and 4.66 MeV; broad structure at 6–8 MeV.
- The deuteron $T_d(\theta)$ band must follow the analytic g.s. kinematic line — if not, suspect a wrong beam energy, B -field sign, or frame convention.

13 Troubleshooting

- “*No raw/reco files found.*” Check `$DATA` and the `filepath` line in the reconstruction macro.
- “*/tmp/pd_kin.root missing.*” `/tmp` clears on reboot — rebuild the cache (Step 3).
- “*Gate file not found.*” Run from the UKF directory so `pid/...json` resolves.
- “*PID macro is slow.*” The IC gate reads the large FRIB files and is opt-in; without it, `pid_pd.C` only reads the small PID branch and is fast.
- “*GUI does not open.*” The interactive macros need X — run with `root -l` (drop `-b`).